

BHARATH NALLAMOTHU

Principal AI Engineer

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PROFESSIONAL SUMMARY

Principal AI Engineer with 11+ years of experience designing and delivering production AI and software platforms across automotive, telecommunications, healthcare, and education. Core contributor to enterprise Generative AI products at Toyota Motor North America spanning multi-agent research, manufacturing troubleshooting, model and tool orchestration, retrieval, memory, and citation-backed workflows. Combines Python, FastAPI, LangGraph, RAG, evaluation, AWS cloud, system design, and full-stack product delivery to build reliable, scalable, and cost-aware AI systems.

TECHNICAL SKILLS

Agentic AI and Evaluation: Python, FastAPI, LangChain, LangGraph, LangSmith, RAGAS, ReAct, agent and tool orchestration, supervisor and hierarchical agents, context engineering, human-in-the-loop workflows
Retrieval and Data: RAG, OpenSearch, FAISS, HNSW, knowledge graphs, lexical, semantic, hybrid and vector search, metadata filtering, retrieval evaluation, Text-to-SQL, PostgreSQL, Redis
AWS and Engineering: Amazon Bedrock, SageMaker, ECS, Lambda, Step Functions, DynamoDB, S3, S3 Vectors, Docker, TypeScript, Node.js, React, REST APIs, GraphQL

PROFESSIONAL EXPERIENCE

Principal AI Engineer | Toyota Motor North America

Feb 2024 - Present

- Build Python and FastAPI services, LangGraph agent workflows, retrieval pipelines, tool integrations, and production user experiences across TMNA Generative AI products; apply context engineering, LangSmith observability, RAGAS evaluation, prompt optimization, retries, fallbacks, and human-in-the-loop controls.
- **GearPal:** Co-developed an agentic RAG troubleshooting assistant with permission-aware OpenSearch routing, hybrid and vector retrieval, metadata filters, API/tool integration, and confidence-based fallback. The team platform supports 20 shops across 7 North American plants and 1,300+ maintenance professionals.
- **ToyotaGPT:** Implemented LangGraph workflows for conditional execution, model and tool selection, structured responses, conversation memory, session restoration, history retrieval, and reusable application integrations.
- **MaterialGPT:** Co-developed multi-agent RAG research across service reports, ANSYS data, technical documentation, and PostgreSQL Text-to-SQL using planning, specialist agents, follow-ups, and citation-backed synthesis. The team platform serves 1,000+ users and reduced initial research discovery from 4-5 days to minutes.
- **BatteryBrainGPT:** Built retrieval and LLM orchestration for a battery engineering assistant connecting technical knowledge sources and APIs for diagnostics, research, and troubleshooting.

Full Stack Developer | Verizon

Oct 2022 - Feb 2024

- Designed TypeScript and Node.js microservices using Express.js, PostgreSQL, REST APIs, GraphQL, Apollo, and NGINX; supported OAuth 2.0, Dockerized AWS deployments, OpenShift, and Jenkins CI/CD.
- Built React and TypeScript applications with Redux, API integrations, reusable components, code reviews, and Jest/React Testing Library tests.

ADDITIONAL EXPERIENCE

- **Novartis** - Sr Software Developer (Oct 2020 - Oct 2022): Delivered React applications backed by Node.js, GraphQL/Apollo, REST APIs, Jest, AWS, and Docker.
- **Study Monitor** - Sr Software Engineer - Lead (May 2018 - Oct 2020): Led React and React Native delivery using Redux, Node.js/LoopBack, MongoDB, AWS, REST APIs, and automated testing.
- **LivBIM** - Software Developer (Apr 2017 - May 2018): Built Angular and Node.js/LoopBack applications, reusable components, WebSocket-driven experiences, and AWS-hosted services.
- **YITSOL** - Software Developer (Sep 2015 - Apr 2017): Developed React, AngularJS, JavaScript, Node.js, and REST-based interfaces deployed with Docker, OpenShift, and AWS.
- **CMC, A TATA Enterprise** - IT Engineer (Jun 2014 - Sep 2015): Built MVC web interfaces using HTML, CSS, JavaScript, jQuery, JSON, and AJAX.

EDUCATION

Bachelor's Degree | Jawaharlal Nehru Technological University